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Ultra-low frequency oscillation analysis considering thermal-hydro power proportion

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ABSTRACT

The Ultra-low Frequency Oscillation (ULFO) has attracted much attention due to its field test capture in China, and it is believed that the ULFO is related to the hydraulic generator. To clarify the cause and innate essence of ULFO, the develop of the ULFO under different thermal-hydro power proportions is investigated. A small-signal model of two-machine system considering different thermal-hydro power proportions is established. Based on the small-signal model, the influence of hydropower proportion on the oscillation characteristics is analyzed in different conditions, and it is found that a higher proportion of hydro power results in a lower oscillation frequency, though the hydropower proportion is not the crucial cause in ULFO. Then Based on the eigenvalue analysis, it is observed that the essential cause of ULFO is the weak grid condition rather than the high hydropower proportion. The system connection strength determines whether the system is prone to ULFO. In addition, the participating factors of the generators' significant state variables are obtained, which indicates that the governor plays a more vital role in ULFO in the case of weak grid connection. To verify the theoretical analysis, the simulations of the two-machine system and six-machine nine-node system are implemented, respectively. The results reveal that system with all thermal power is still prone to ULFO when the connection is weak. Conversely, when the connection is strong, the ULFO does not occur even if all the generators are hydraulic. Furthermore, in a weak grid condition, the oscillation frequency and damping ratio decrease with increasing hydropower proportion, which indicates that a high hydropower proportion deteriorates the ULFO problem.

1. Introduction

Ultra-low frequency oscillation (ULFO) refers to the oscillation with an extremely low oscillation frequency frequencies, which is monitored more than once in the actual project tests in China. The oscillation frequency of ULFO is neglected and lower than 0.1 Hz. Actually, the occurrence of ULFO can be tracked as early as 1966, when the US Northwest Pacific system first reported the ULFO with oscillation frequency of 0.042 Hz–0.058 Hz [1]. After that, serval other ULFO cases with similar frequency have been successively investigated in Turkey and Colombia [2,3]. In contrast to widely studied low frequency oscillation (LFO), ULFO does not appear often and can be categorized as mechanical oscillation, since the oscillation degree of mechanical torque is more prominent than electromagnetic torque [4], yet, the intrinsic factor has not yet been elucidated.

In the last few years, the ULFO has reappeared in Chinese actual project. The oscillations of 0.1 Hz and 0.07 Hz were reported when the field test of isolated operation of Tianguang and Jinsu HVDC transmission line were conducted, respectively. And the corresponding oscillation amplitudes were ±0.23 Hz and ±0.26 Hz, respectively [5]. Subsequently, the asynchronous interconnection filed test between in Southern China Power Grid experienced 0.05 Hz ULFO and lasted for half an hour in 2016, which was with high proportion of hydropower [6,7].

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To clarify the ULFO theory, several studies have analyzed the ULFO mechanism based on linear methods. The damping torque was widely utilized since the influence of each part can be intuitively analyzed [8,9]. The hydraulic governor plays the dominant role leading to ULFO in the high-ratio hydropower system due to that the unexpected negative damping [10,11]. Furthermore, the damping torques provided by the hydraulic governor under various operation modes and PID parameters were derived, and it was concluded that the water hammer effect time constant T_W was another crucial parameter that significantly influenced ULFO in [12,13]. On the other hand, according to the research based on eigenvalue analysis, the ULFO phenomenon was also found to be closely related to the hydraulic governor [14–17], which was consistent with aforementioned findings. By nonlinear method and taking coupling characteristics into consideration, the stability limit cycles with different T_W were analyzed, which showed the stability was deteriorated with increasing T_W [18,19]. Specifically, by the Routh criterion, the ratio of the proportional gain to integral gain of the hydraulic governor would cause ULFO if it was small enough [20,21]. Besides, the large phase lag and the positive energy provided by the hydropower were also declared to be the causes of ULFO [22,23].

It can be seen from the above analysis that the negative damping effect in the ultra-low frequency band caused by the hydraulic generator is considered to be the inner source of ULFO. Yet, the practical system which is not asynchronously connected is not prone to ULFO, and in the system with 100 % hydropower, it was only risky of power oscillations caused by the bulb turbine torque disturbance [24,25]. Moreover, the system was prone to LFO (local mode) on account of the developed rapidity of governing control in hydropower plants (HPPs) [26,27]. Furthermore, although several ULFOs occur in the system with high hydropower proportions, the issue might also occur in systems with 100 % thermal power [28]. Previous studies on ULFO didn't identify the causes and occurrence circumstances of ULFO, because the hydraulic governor was in operation for a long time. And the study results are not sufficient to explain the unusual scene that ULFO generally occurs in an asynchronous operated condition.

To fill in the gap and explain the scenario of ULFO issue, this paper investigates the specific influence of specific hydro-thermal proportion and the analyses mechanism of strength of the system in the occurrence of ULFO. First, a small-signal model of two-machine system is derived to analyze the development law of different hydro-thermal power proportions on oscillation modes. After that, to find out the key factors that make the system prone to ULFO, the influence of connection strength on oscillation characteristics is investigated through theoretical and simulation models. Moreover, based on the eigenvalue analysis, it is observed that the weak grid condition rather than the high hydropower proportion makes the system risky of ULFO. Under this precondition, the negative damping effect caused by the hydraulic governor is more prominent with increasing hydropower proportion. Furthermore, the hydraulic governor dominates the performance of ULFO according to the evaluation results of participating factors.

The remainder of this paper is organized as follows. Section 2 derives the small-signal analysis model of the two-machine system, containing speeding governing system, prime mover and exciter of different types of units. Section 3 studies the influence of the connection strength on oscillation characteristics by obtained eigenvalue and the crucial variables causing ULFO are identified. Section 4 investigates the participating factors of all state variables to reveal their relationship with oscillation characteristics. Then the simulation results are conducted and the correctness are verified in Section 5. Finally, main conclusions are draw in the last section.

2. Theoretical model with hydropower and thermal power

To clarify the effect of proportion of hydro and thermal power on ULFO, the two-machine system with different hydropower proportions is established, as depicted in Fig. 1. And the theoretical model of this system is investigated in this section.

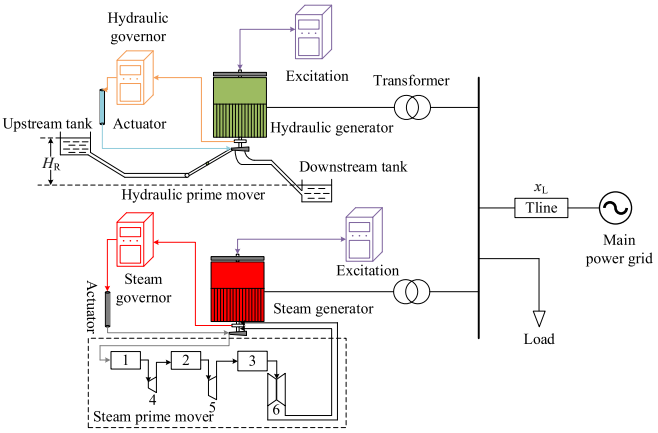


Fig. 1. Two-generator system; 1-steam chest; 2-reheater; 3-crossover; 4-high pressure cylinder; 5-medium pressure cylinder; 6-low pressure cylinder.

2.1. Synchronous machine

The synchronous machine can be expressed as (1)–(4), there have four identical variables, which are $\Delta\omega$, $\Delta\delta$, $\Delta E''_q$, and $\Delta E'_q$. They can be derived as

$$\Delta\dot{\delta} = \omega_0\Delta\omega \tag{1}$$

$$\Delta\dot{\omega} = \frac{(\Delta M_m - \Delta M_e)}{T_J} \tag{2}$$

$$T'_{d0}\Delta\dot{E}'_q = \Delta E_f - \frac{x_d - x_l}{x'_d - x_l}\Delta E'_q + \frac{x_d - x'_d}{x'_d - x_l}\Delta E''_q - \frac{(x_d - x'_d)(x''_d - x_l)}{(x'_d - x_l)}\Delta i_d \tag{3}$$

$$T''_{d0}\Delta\dot{E}''_q = \frac{x''_d - x_l}{x'_d - x_l}T''_{d0}\Delta\dot{E}'_q - \Delta E''_q + \Delta E'_q - (x'_d - x''_d)\Delta i_d \tag{4}$$

where δ is the angle between the q-axis of the i-th generator and the common D-axis. ω is the speed of generator, and ω_0 is the rated value. M_m and M_e are the generator's mechanical and electromagnetic torque, respectively. T_J is the inertia coefficient of the rotor. T'_{d0} and T''_{d0} are the d-axis transient and sub-transient time constant, respectively. E'_q and E''_q are the q-axis transient voltage and sub-transient voltage, respectively. E_f is the field voltage, and the subscripts d and q denote components of d-axis and q-axis, x_d is the d-axis reactance. x_1 is the stator reactance. x'_d and x''_d are transient and sub-transient reactance, respectively. i_d and i_q are the current component. Δ is the linearization symbol.

As the two types of generator have different windings, the expressions of the remaining state variables are different from each other.

The hydraulic generator's d-axis sub-transient voltage, i.e., $\Delta E''_d$, is expressed by (5), where x_q is the q-axis reactance and x''_q is the sub-transient reactance, T''_{q0} is the sub-transient time.

$$T''_{q0}\Delta\dot{E}''_d = -\Delta E''_d + (x_q - x''_d)\Delta i_q \tag{5}$$

The steam generator's d-axis sub-transient voltage is expressed by (6).

$$T''_{q0}\Delta\dot{E}''_d = \frac{x''_q - x_l}{x'_q - x_l}T''_{q0}\Delta\dot{E}'_d - \Delta E''_d + \Delta E'_d + (x'_q - x''_q)\Delta i_q \tag{6}$$

where E'_d is the steam generator's d-axis transient voltage.

The steam generator's d-axis transient voltage is represented by (7).

$$T'_{q0}\Delta\dot{E}'_{d} = -\frac{x_q - x_1}{x'_q - x_1}\Delta E'_q + \frac{x_q - x'_q}{x'_q - x_1}\Delta E'_d + \frac{(x_q - x'_q)(x''_q - x_1)}{x'_q - x_1}\Delta i_q \tag{7}$$

where T'_{q0} is the q-axis transient time.

2.2. Exciter

Besides, the model of exciter is shown in Fig. 2, which is based on IEEE standard [29,30].

In Fig. 2, U_{ref} is the referenced voltage. K_A and K_E are the proportional coefficient, T_A, T_E and T_F are the time constant of voltage regulator, exciter and feedback loop, respectively. By choosing the output of each part U_R, U_F and E_f as the state variables, the exciter can be described as follows.

$$\Delta\dot{E}_f = \frac{1}{T_E}\Delta U_R - \frac{K_E}{T_E}\Delta E_f \tag{8}$$

$$\Delta\dot{U}_F = \frac{K_F}{T_F}\Delta\dot{E}_f - \frac{1}{T_F}\Delta U_F \tag{9}$$

$$\Delta\dot{U}_R = -\frac{1}{T_A}\Delta U_R - \frac{K_A}{T_A}\Delta U_t - \frac{K_A}{T_A}\Delta U_F \tag{10}$$

2.3. Hydraulic governing system

The governors of the hydraulic and steam generators are absolutely different. The control system of the hydraulic governor is displayed by Fig. 3, where K_{P1}, K_{d1}, K_{i1} are the PID control parameters of governing part, b_p is the permanent slip coefficient. The actuator and hydraulic prime mover are illustrated by Fig. 4, where K_P, K_d, K_i are the control parameters of hydraulic actuator. T_f denotes the inertial link parameter of actuator, and T_w describes water hammer effect.

Similarly, choosing the outputs integration elements as state variables in Fig. 3 and Fig. 4, the expressions of the hydraulic governing system can be written as follows.

$$\Delta\dot{G}_2 = -\frac{K_{i1}(\Delta\omega_1 + b_P\Delta G_1 + b_P\Delta G_2)}{K_{P1}b_P + 1} \tag{11}$$

$$\Delta\dot{G}_1 = -\frac{\left(\Delta G_1 + K_{d1}\Delta\dot{\omega}_1 + K_{d1}b_P\Delta\dot{G}_2 + K_{P1}b_P\Delta G_1\right)}{T_{d1} + K_{d1}b_P + K_{P1}T_{d1}b_P} \tag{12}$$

$$\Delta\dot{G}_3 = -\frac{K_i[\Delta G_4 - (\Delta G_1 + \Delta G_2 - K_{P1}\Delta\omega_1)]}{K_{P1}b_P + 1} \tag{13}$$

$$\Delta\dot{G}_4 = \frac{(\Delta P_{GV} - \Delta G_4)}{T_f} \tag{14}$$

$$\Delta P_{CV} = \frac{(\Delta G_1 + \Delta G_2 - K_{P1}\Delta\omega_1)}{(1 + K_{P1}b_P)} \tag{15}$$

$$\Delta\dot{P}_{GV} = K_P(\Delta P_{CV} - \Delta G_4) + K_d(\Delta\dot{P}_{CV} - \Delta\dot{G}_4) + \Delta G_3 \tag{16}$$

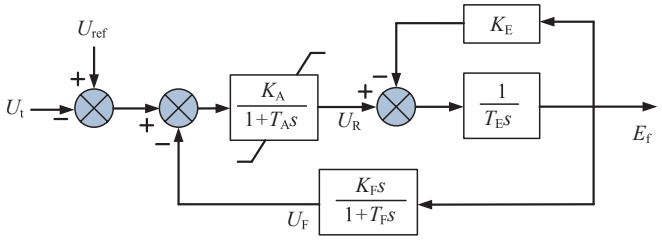


Fig. 2. Exciter model.

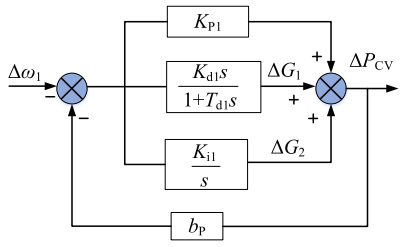


Fig. 3. Hydraulic governor.

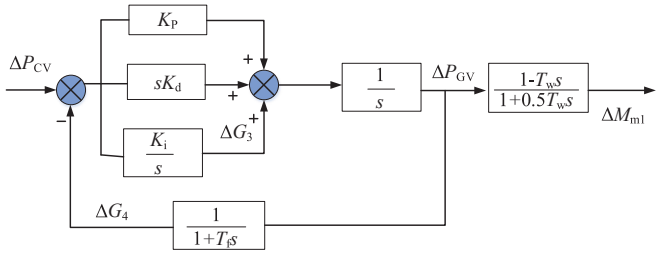


Fig. 4. Hydraulic actuator and prime mover.

$$\Delta\dot{M}_{m1} = \frac{1}{0.5T_W}(\Delta P_{GV} - T_W\Delta\dot{P}_{GV} - \Delta M_{m1}) \tag{17}$$

2.4. Steam governor

As for steam unit, the models of governor, actuator, and prime mover are depicted as well, where K_R is the magnification. T_{lead} and T_{lag} are the lag and lead time constants of the speed relay, respectively. T_{SM}, T_{CH}, and T_{RH} are the gate servo, steam chest, and reheater time constants, respectively. T_{CO} is the switching time of reheater. FHP, FIP, and FLP are the output power proportions of high, medium and low pressure cylinders, respectively.

In a similar way, the state variables are shown in Fig. 5 and Fig. 6, the steam governing system can be obtained by

$$\Delta\dot{x}_1 = \frac{-K_R\Delta\omega - K_RT_{lead}\Delta\dot{\omega} - \Delta x_1}{T_{lag}} \tag{18}$$

$$\Delta\dot{x}_2 = \frac{1}{T_{SM}}(-\Delta x_1 - \Delta x_2) \tag{19}$$

$$\Delta\dot{x}_3 = \frac{1}{T_{CH}}(-\Delta x_2 - \Delta x_3) \tag{20}$$

$$\Delta\dot{x}_4 = \frac{1}{T_{RH}}(\Delta x_3 - \Delta x_4) \tag{21}$$

$$\Delta\dot{x}_5 = \frac{1}{T_{CO}}(\Delta x_4 - \Delta x_5) \tag{22}$$

The modified Philips-Heffron model of the two-generator system are established in [31], the non-state variables in these exprssions can be expressed by the state variables. Consequently, combining (1)–(22), the integral small-signal model of two-machine system is derived as (23). And Δx is the matrix containing all the state variables.

$$\Delta\dot{x} = A\Delta x \tag{23}$$

3. Cause of ULFO considering different thermal-hydro power proportions

It is well-known that ULFO issue is mainly caused by hydropower because of its unique water hammer effect. Thus, this section will

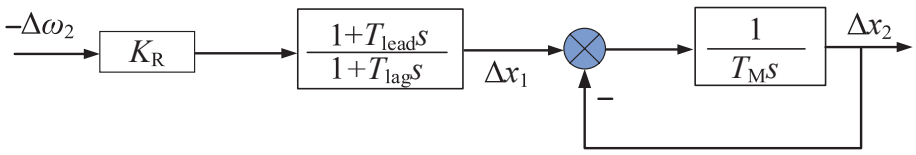


Fig. 5. Steam governor and actuator.

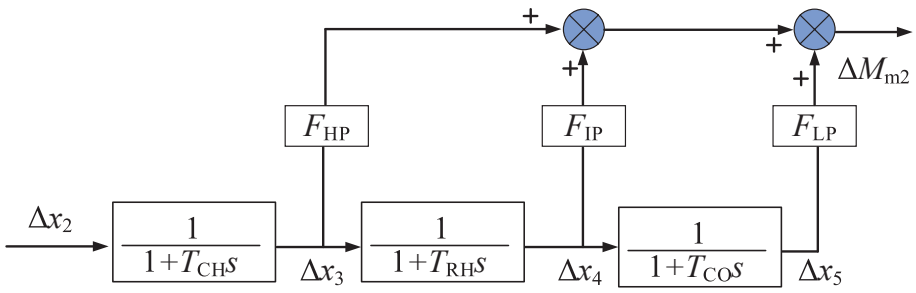


Fig. 6. Steam prime mover.

examine its correctness based on the established small-signal models considering different thermal-hydro power proportions. Further, the parameters used in the investigation are presented in the supporting document, and it is provided by the practical project. The based power is 3000 MW. As aforementioned, the strength of system should also be taken into account.

3.1. Cause of ULFO

To investigate the cause of ULFO as well as the influence of different thermal-hydro power proportions, the outputs of the hydraulic and steam generators are adjusted to conduct various hydro-thermal proportion. Thus, three typical situations are selected for the preliminary study as shown in Table 1. In addition, the system strength is reflected by the reactance of transmission line x_L. And the green area indicates ULFO modes.

The changing situations of the root locus with different thermal-hydro proportions are correspondingly presented in Figs. 7–9 when x_L changes of cases 1, 2, and 3, respectively. The root locus diagrams with the variations of other parameters are given in the supporting document.

According to the eigenvalue analysis results, all three cases have the risk of ULFO if the transmission line reactance x_L is large enough, which intends a weak connected sending system. As shown in Fig. 7, even if the generators are all steam units, the system is still prone to ULFO under a weak grid condition. Meanwhile, Fig. 9 also indicates that even if the generators are all hydraulic generators, the risk of ULFO under a strong grid condition is considerably low. The same conclusion can be drawn from Fig. 8. Thus, it can be concluded that the connection strength is the crucial external factor that leads to ULFO rather than the hydropower proportion.

Nonetheless, when the connection is weak, it is observed that when the hydropower proportion increases, the dominant eigenvalue shifts towards the imaginary axis and eventually becomes positive, in other words, the damping ratio decreases. Therefore, the system is more prone to instability with a higher hydropower proportion in weak grid condition.

Table 1
The typical cases for study.

Case	Hydropower proportion	Thermal power proportion
1	0 %	100 %
2	50 %	50 %
3	100 %	0 %

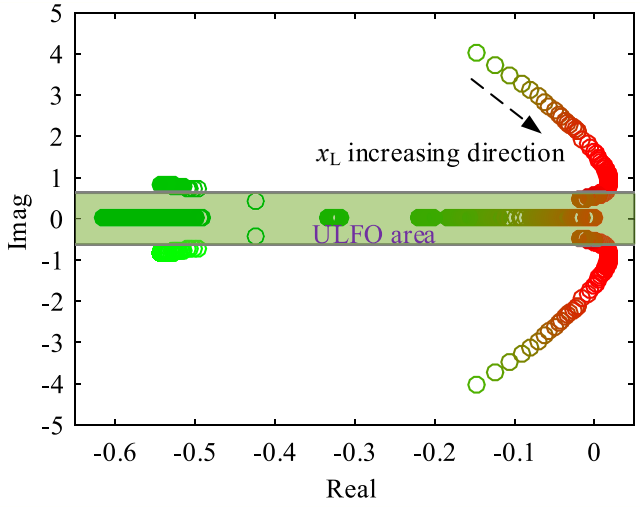


Fig. 7. Root locus of case 1 (all thermal power) when system strength changes.

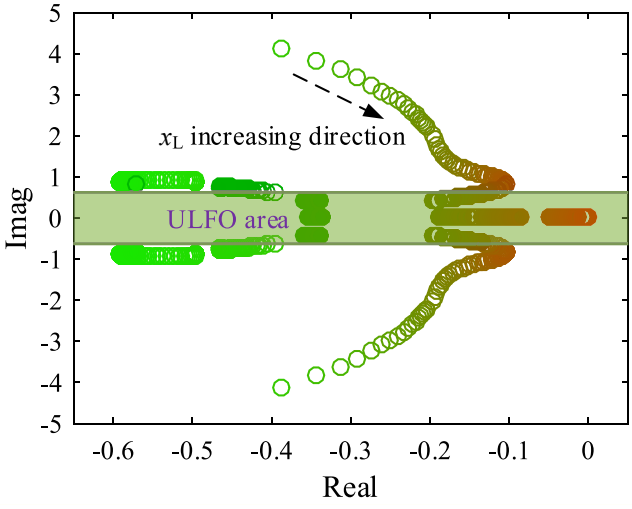


Fig. 8. Root locus of case 2 (50 % thermal power and 50 % hydraulic power) when system strength changes.

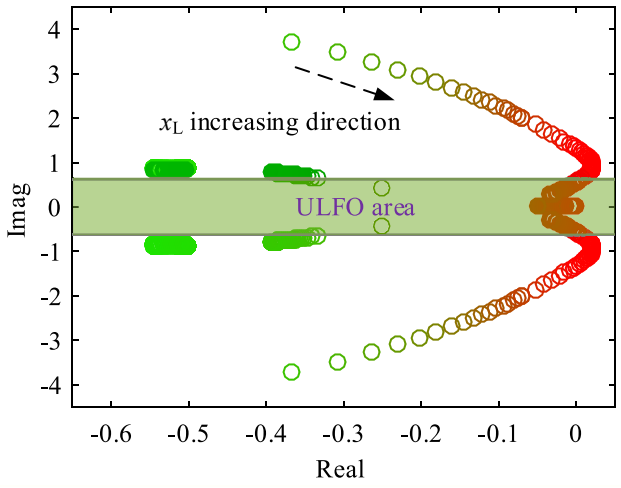


Fig. 9. Root locus of case 3 (all hydraulic power) when system strength changes.

3.2. Influence of hydropower proportion on dynamic characteristics

Since the previous study only investigated three typical cases, this section evaluates the influence of hydropower proportion under different line reactances to further confirm the main cause of ULFO.

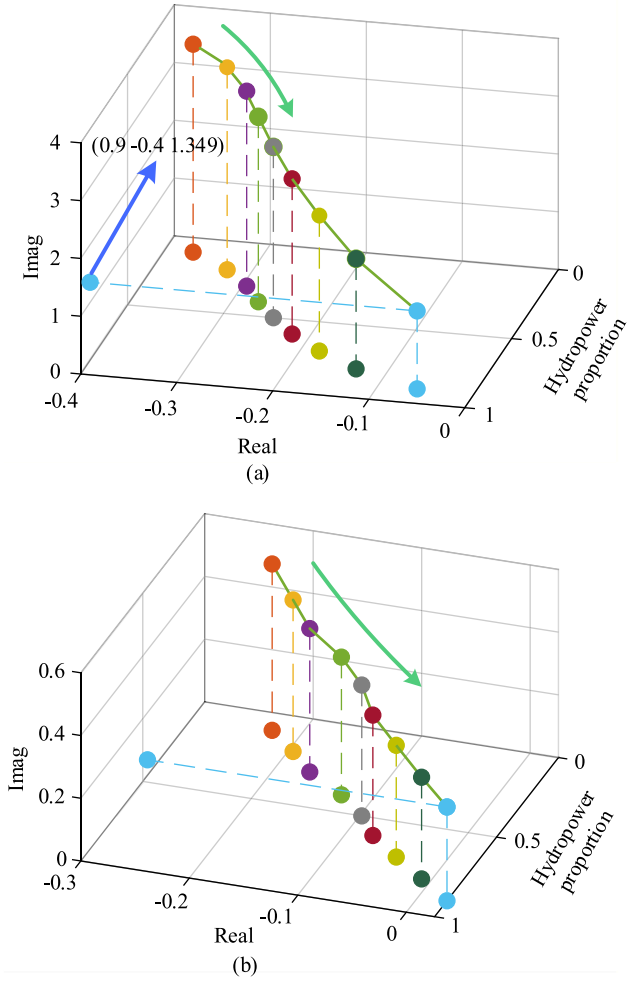


Fig. 10. Dominant eigenvalues with different hydropower proportions (a) in a strong grid, (b) in a weak grid.

Fig. 10 presents variations of the dominant eigenvalue, and the dominant eigenvalue in the two-dimensional plane is depicted by Fig. 11. The axis denoted by "Real" and "Imag" represent the real and imaginary elements of the eigenvalues, respectively. Meanwhile, the real part reflect the damping level and the imaginary parts reflect the oscillation frequency of the system.

From Fig. 10(a), it is obvious that the imaginary part of the eigenvalue reaches the minimum value of 1.349 when the hydropower proportion reaches to 90 %, and it is beyond ultra-low frequency band. On the contrary, in Fig. 10 (b), the system is prone to ULFO under a weak grid condition even if the hydropower proportion is only 10 %. And this tendency presented by Fig. 11 is more intuitive.

In Fig. 11, it is obtained that the damping level of the system would weaken with the increase of hydropower proportion no matter the grid is strong or weak. However, in strong grid condition, the imaginary part of the dominant eigenvalue does not fall into the ultra-low frequency band even if the grid is with a high hydropower proportion. Moreover, the damp of system is positive due to the negative real part. Conversely, in the weak grid condition, the system is prone to ULFO even under a low hydropower proportion, and the system may be even unstable as the hydropower proportion increases.

Thus, it is rationally revealed that the connection strength is the dominant factor which leads to ULFO, and the system with a high hydropower proportion is more prone to instability influenced by the peculiar water hammer under the precondition of a weak grid condition.

4. Impacts of controller parameters on ULFO in weak grid conditions

As aforementioned, it can be concluded that the weak condition of interconnection determines the occurrence of ULFO. In this section, the detailed impact of the generators' inner variables on ULFO is investigated when the system is weakly connected.

4.1. Participating factors of the state variables under a weak grid condition

To further clarify the parameters' influence, the participating factors are utilized and derived. The calculated eigenvalue λ_i should satisfy equations (24) and (25), where ϕ_i and ψ_i are the right eigenvector and left eigenvector of matrix A, respectively.

$$\boldsymbol{A\phi}_i = \lambda_i \boldsymbol{\phi}_i \tag{24}$$

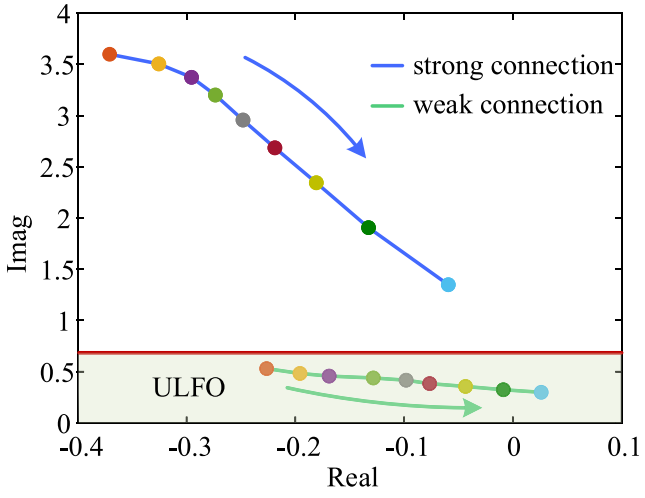


Fig. 11. Dominant eigenvalues with different hydropower proportions under different grid conditions.

$$\boldsymbol{\psi}_i \mathbf{A} = \lambda_i \boldsymbol{\psi}_i \tag{25}$$

Then the participating factors are delivered by the kth terms of corresponding eigenvector in (26), p_{ki} can evaluate the participation degree of state variable.

$$p_{ki} = \phi_{ki} \psi_{ik} \tag{26}$$

Thus, considering weak grid connection, the participating factors of different variables are shown in Figs. 12–14, where the system has different thermal-hydro power proportions. As shown in Fig. 12, the participating factors of the variables $\Delta\omega$ and angle $\Delta\delta$ are larger than other variables, in other words, the exciter has little effect on ULFO. Similarly, based on the results shown in Fig. 13, the state variables related to generator windings have little effect on ULFO. In Fig. 14, the participating factors of Δx_4, ΔG_2, ΔP_{GV}, ΔM_{m1}, and Δx_3 are also large enough, where Δx_4 is the largest. Thus, it indicates that the hydraulic and steam governors always participate in ULFO. Furthermore, the hydraulic governor performs more deeply on ULFO with the increase of hydropower proportion.

4.2. Influence of controller parameters on ULFO under a weak grid condition

To further validate the conclusion that both the hydraulic and steam generators participate in ULFO, the variations of the eigenvalue under different governor parameters are presented by Fig. 15, where the hydropower proportion is set as 80 %.

From Fig. 15 (a), it is indicated that the damping ratio decreases with increasing T_W as well as oscillation frequency, and the system becomes unstable when T_W increases sufficiently. Especially, when the parameters T_{SM}, T_{CH} and T_{RH} increase, the situations are the same as that when T_W changes.

4.3. Water hammer effect in the system with high hydro-power proportion

Due to the fact that system is prone to ULFO under a weak grid condition, and the water hammer of hydraulic generator deteriorates the system stability. The influence of the water hammer effect on the system stability is elaborated as well.

As shown in Fig. 16, the system resembles a non-minimum phase system due to the water hammer effect. Specifically, when the opening valve μ increases abruptly, the water flow into the hydraulic turbine increases. However, the speed of the water flow in the lower section of the water diversion pipeline cannot respond in time. Therefore, the water pressure entering the turbine decreases, and the instantaneous power input to the hydraulic turbine will not increase but decrease. As a result, the system exhibits the characteristics of a non-minimum phase system.

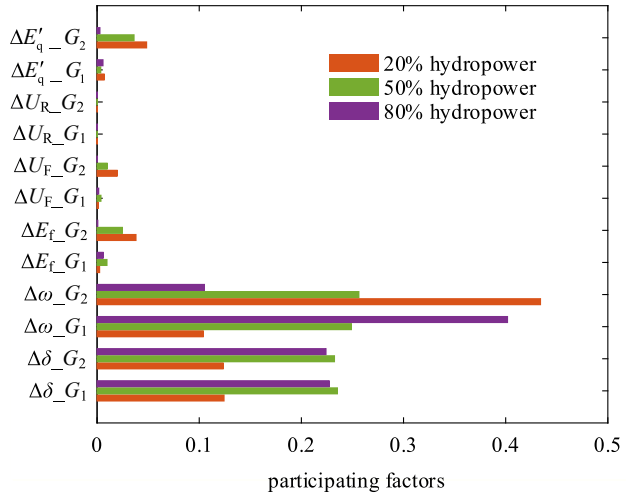


Fig. 12. State variables' participating factors of the generators and exciter.

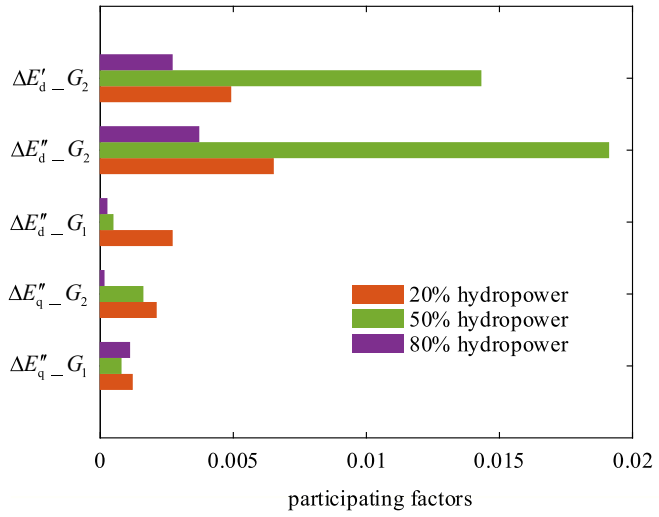


Fig. 13. State variables' participating factors of the generators' windings.

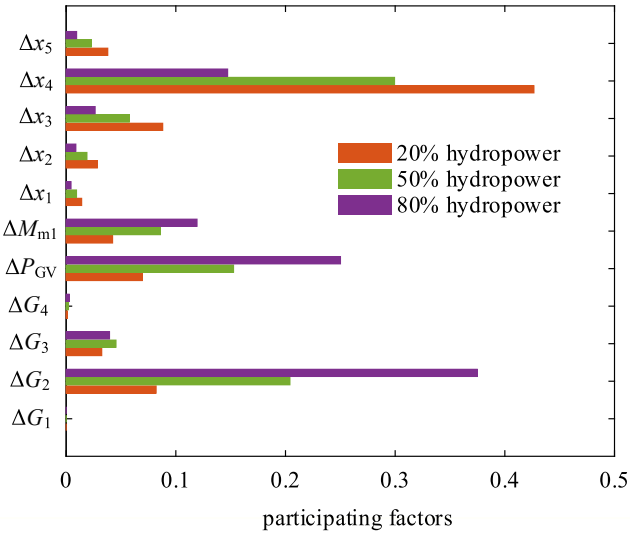


Fig. 14. State variables' participating factors of the governor and turbine.

To further reveal the influence of the water hammer effect T_W on ULFO, the damping torque caused by hydraulic governor is expressed in (27), where the detailed expressions of the coefficients are given in the supplementary document.

$$K_D = \frac{K_{g1}K_{g3} + K_{g2}K_{g4}}{K_{g1}^2 + K_{g2}^2} \tag{27}$$

The damping torque provided by hydraulic and steam governing systems is depicted by Fig. 17 and Fig. 18, respectively. From Fig. 17, it is indicated that the negative damping characteristics are more prominent in the band of 0–0.1 Hz with increasing Tw. On the contrary, Fig. 18 shows that the damping provided by steam governor remains positive in the band below 01 Hz. Therefore, in high-proportion hydropower system, there is a higher risk of instability of ULFO caused by water hammer effect.

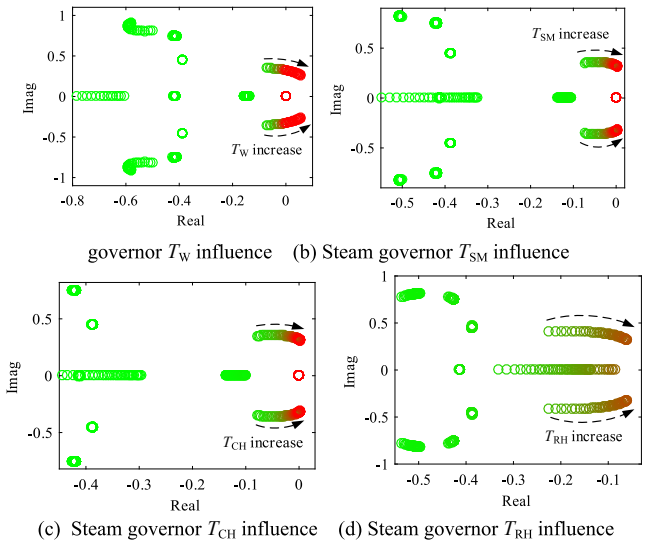


Fig. 15. Governor parameters' influence on ULFO.

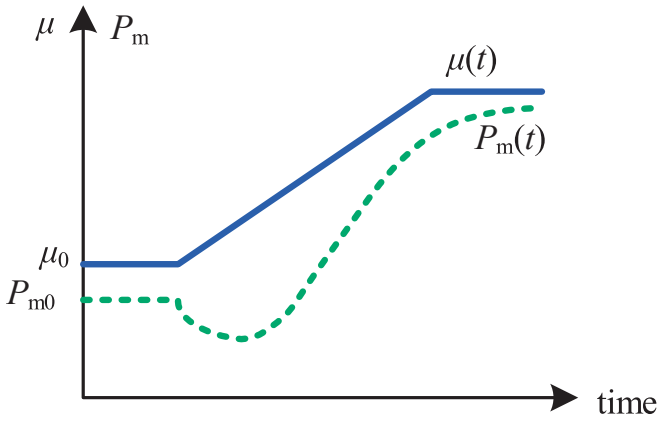


Fig. 16. Water hammer effect.

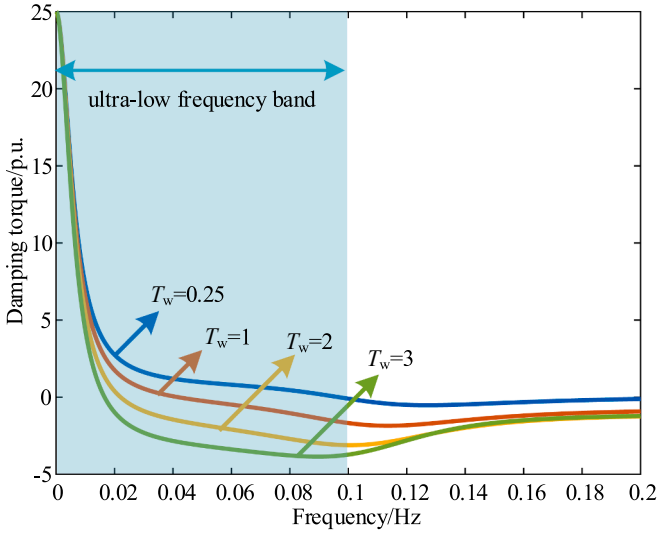


Fig. 17. Damping torque provided by the hydraulic governor.

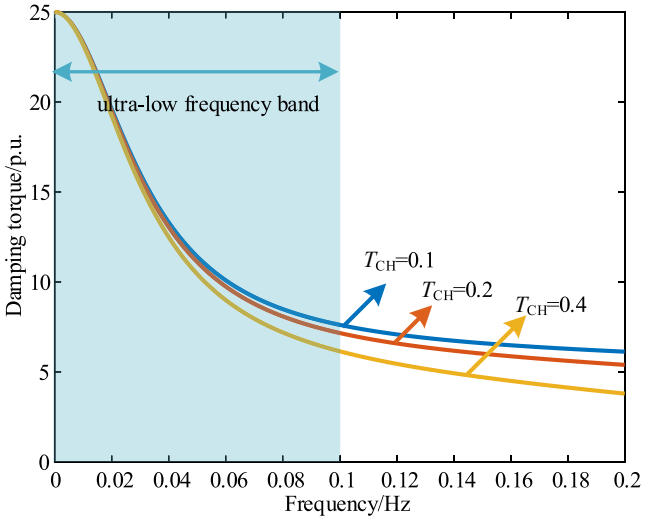


Fig. 18. Damping torque provided by the steam governor.

5. Simulation verification

In this section, the simulations based on the two-machine system in Fig. 1 are implemented in PSCAD software. Meanwhile, the simulations of a six-machine model in PSASP software are also presented for further verification.

5.1. Simulations based on the two-machine system

1) Verification of the influence of connection strength

The simulation results with different thermal-hydro power proportions under different grid conditions are presented in Figs. 19–21. As shown in Fig. 19, the oscillation frequency will fall into the ultra-low frequency band with the increase of transmission line reactance even the thermal power accounts for 100 %, which is consistent with the conclusion drawn from Fig. 7. Similarly, the results illustrated by Fig. 20 validates the theoretical analysis under 50 % hydropower proportion, which is depicted by Fig. 8. Especially, even when the hydropower proportion reaches to 100 % as shown in Fig. 22, the oscillation mode is

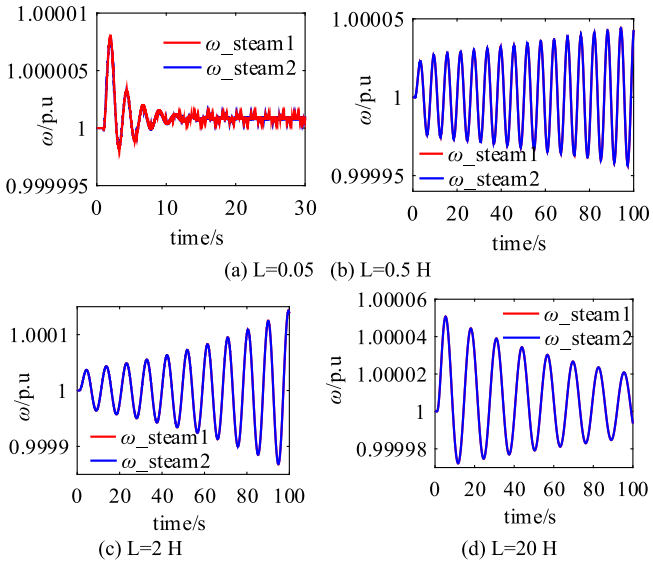


Fig. 19. Influence of system strength on the oscillation characteristics in all thermal power situation.

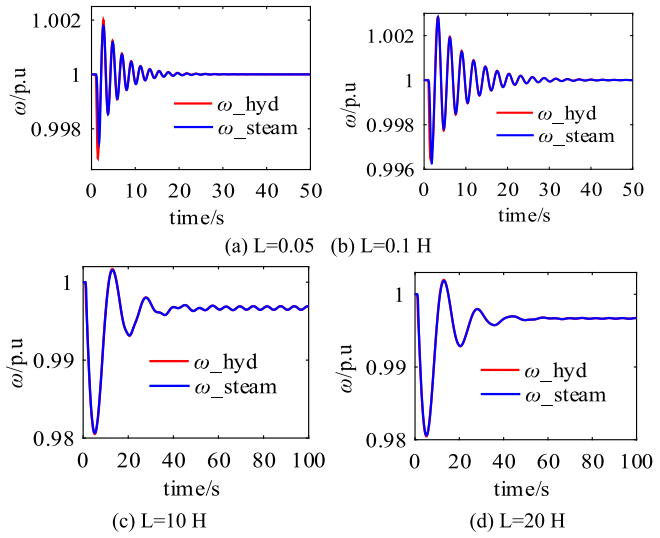


Fig. 20. Influence of system strength on the oscillation characteristics in half thermal and half hydraulic power situation.

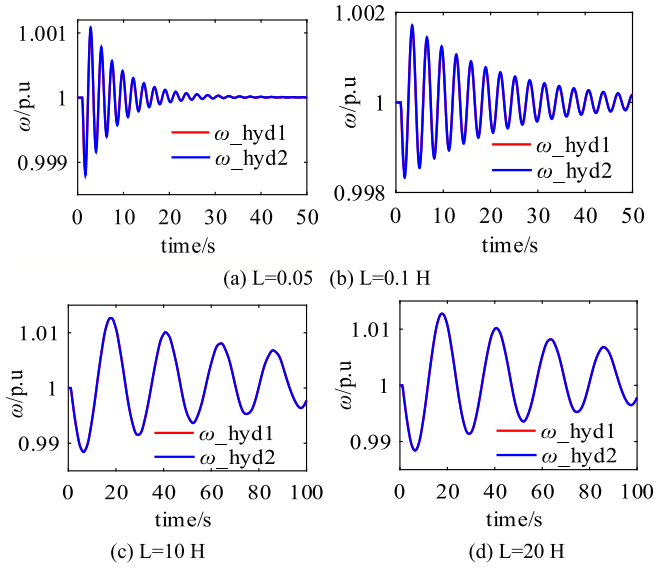


Fig. 21. Influence of system strength on the oscillation characteristics as hydraulic power accounts for 100 %.

not ULFO when x_L is small. It is in alignment with the conclusion drawn from Fig. 9, i.e., the cause of ULFO is the system strength rather than the high hydropower proportion.

Combining the eigenvalues and simulations, it is presented that ULFO occurs in the weak grid condition regardless of the hydropower proportion, that is the reason of that the ULFO can be obtained in asynchronous interconnected grid or islanding hydro-dominated system. Moreover, the LFO occurs when the system is in a strong grid condition. Meanwhile, the LFO in Fig. 1 belongs to the inter-area oscillation as it is the oscillation of the two generators relative to the infinite bus. The oscillation frequency is approximately 0.3 Hz.

2) Influence of hydropower proportion on the oscillation characteristics

To verify the influence of hydropower proportion on oscillation characteristics in different situations, Fig. 22 and Fig. 23 show the simulation results with the increase of hydropower proportion when the system is in a strong and weak connection condition, respectively.

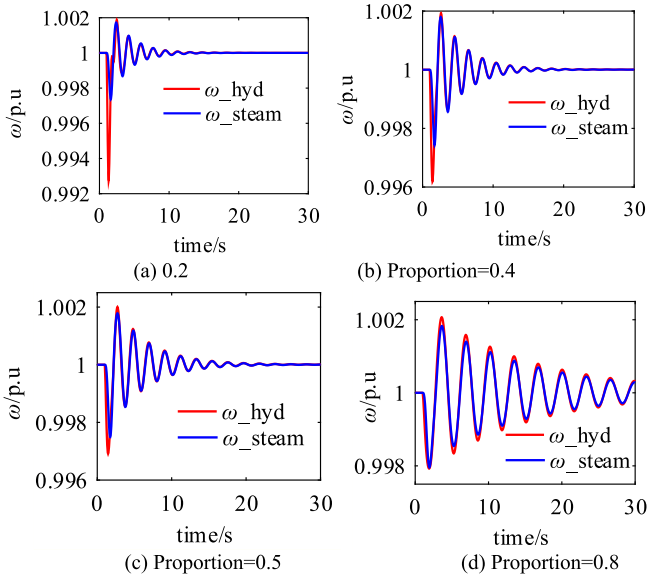


Fig. 22. Influence of hydropower proportion on the oscillation characteristic with strong connection.

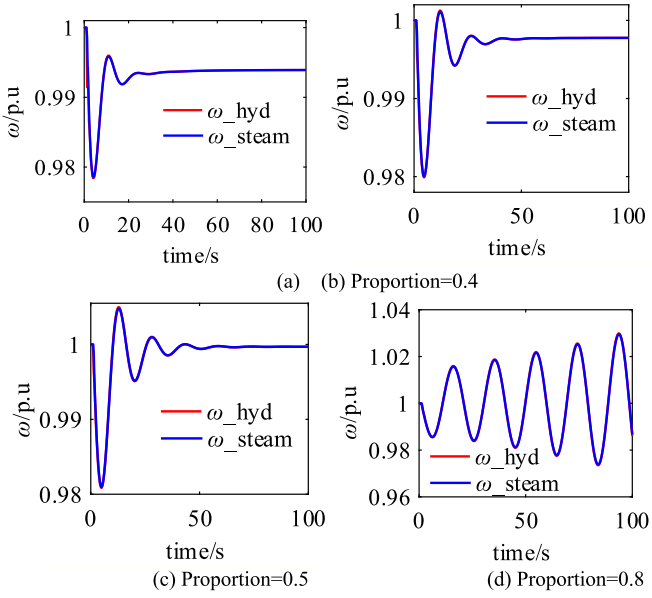


Fig. 23. Influence of hydropower proportion on the oscillation characteristic under a weak grid condition.

As shown in Fig. 22, under the strong grid connection, the grid will not be prone to ULFO even the system is with a high proportion of hydropower, which is consistent with the phenomenon presented by Fig. 10(a). It can be concluded from Fig. 23 that the oscillation mode is ULFO when the system is in a weak grid condition, and the system loses its stability with the increase of hydropower proportion, which is consistent with the conclusion obtained from Fig. 10(b).

Therefore, the above simulation results further verify that the connection strength is the dominant factor that makes the system prone to ULFO. And the simulations also indicate that the increase of the hydropower proportion destabilizes the system, which is consistent with the theoretical analysis illustrated by Fig. 10.

5.2. Simulations based on the six-machine system

This section further validates the proposed theory through a established multiple-machine simulation system. As illustrated in Fig. 24, G1 and G5 are the hydraulic generators, and G_2, G_4, and G_6 are the steam generators.

1) Influence of hydropower proportion on the oscillation characteristics

Similarly, the simulations of six-machine system as shown in Fig. 24 with different connection strengths and thermal-hydro power proportions are conducted as well, which are given in Figs. 25–28, The conclusions are the same as two-machine system, which further confirms that weak grid condition is the main cause of ULFO.

2) Influence of controller parameters on the oscillation characteristics under the weak grid condition

The simulations with different controller parameters based on the six-machine system in a weak grid condition are given in Fig. 28. The results are in accordance with the theoretical analysis in Section 4, i.e., the damping ratio and oscillation frequency decrease with increasing T_W, T_{SM}, T_{CH}, and T_{RH}.

5.3. Comparisons between LFO and ULFO

From the integrated results above, it can be obtained that with the increase of external reactance x_L, the oscillation turns from LFO model to ULFO mode. The essential difference between LFO and ULFO under different connection strengths is the influence on electromagnetic torque T_e and mechanical torque T_m. The comparison between T_e and T_m when the thermal power accounts for 100 % reveals the intrinsic difference between LFO and ULFO.

When x_L is set to 0.02H, the rotor speed is illustrated by Fig. 29(a), and the mechanical torque T_m and electromagnetic torques T_e are depicted by Fig. 29(b). In Fig. 29(a), the oscillation mode belongs to LFO, and the oscillation amplitude of T_e is larger than that of T_m when LFO occurs, as shown in Fig. 29(b). Consequently, the variation of T_m can be neglected when LFO occurs, thereby, the electromagnetic oscillation circuit can be depicted by Fig. 30.

Similarly, ULFO occurs as x_L is set to 20H, and the rotor speed is given by Fig. 31(a). As shown in Fig. 31(b), the variation of T_m is larger than that of T_e, which further demonstrates that the ULFO is highly relative to the mechanical part in the system, i.e., the governor. Therefore, unlike LFO, the influence of T_m cannot be neglected when analyzing ULFO issues as the variation of T_m plays a dominant role.

The above analysis indicates that the dominant torque changes from electromagnetic torque to mechanical torque with the increases of x_L,

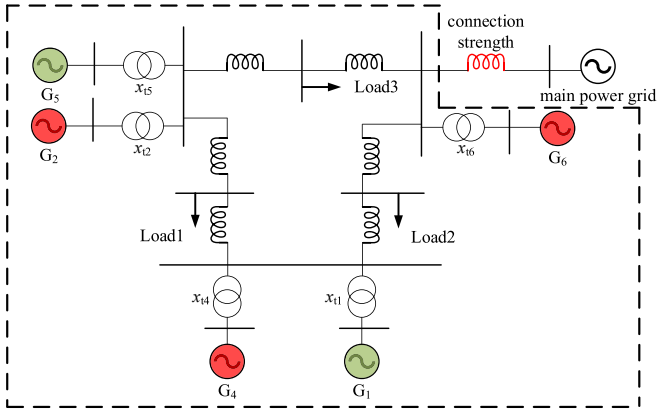


Fig. 24. Six-machine system.

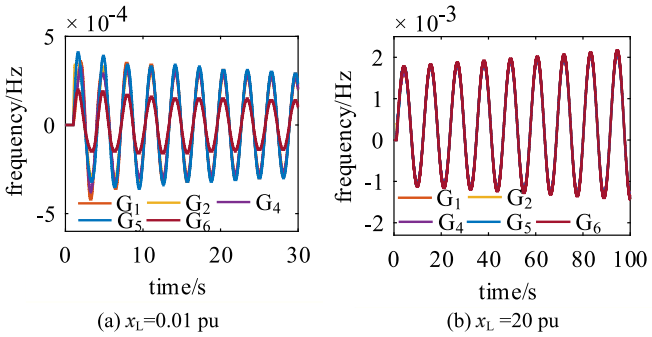


Fig. 25. Influence of connection strength on the oscillation characteristics in all thermal power situation.

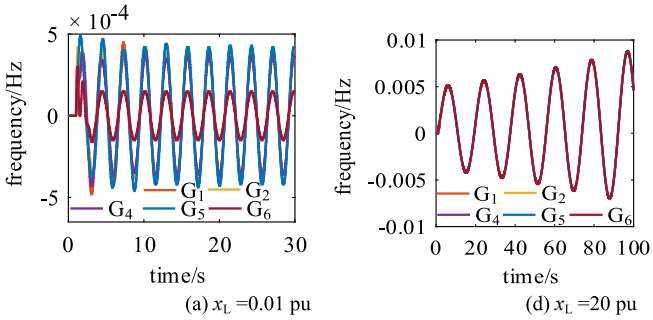


Fig. 26. Influence of connection strength on the oscillation characteristics in half thermal power and half hydropower situation.

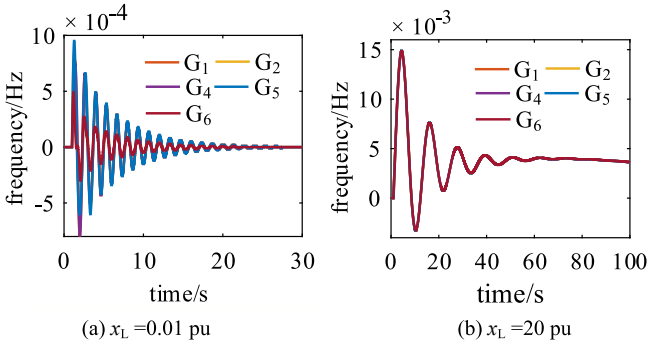


Fig. 27. Influence of connection strength on the oscillation characteristics as hydropower accounts for 100 %.

which also reflects the difference between LFO and ULFO. Therefore, the LFO and ULFO are classified as electromagnetic oscillation and mechanical oscillation, respectively.

6. Conclusion

This study focuses on ULFO issue and conducts further analysis on its occur scenes and precondition, the following significant conclusions are drawn.

(1) The crucial factor leading to ULFO is the connection strength between the sending generator and the main power grid. The system with all thermal power is still prone to ULFO when the connection is weak. On the contrary, when the connection is strong, ULFO does not occur even if all the generators are hydraulic.

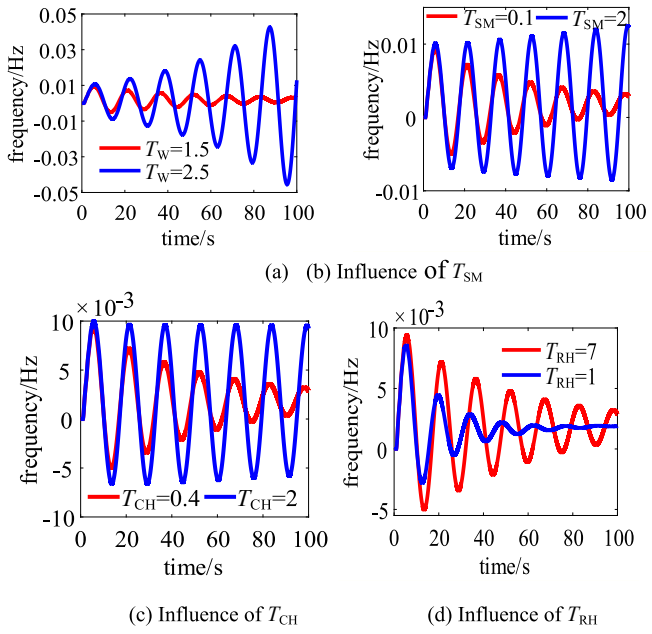


Fig. 28. The influence of controller parameters under the weak gird condition.

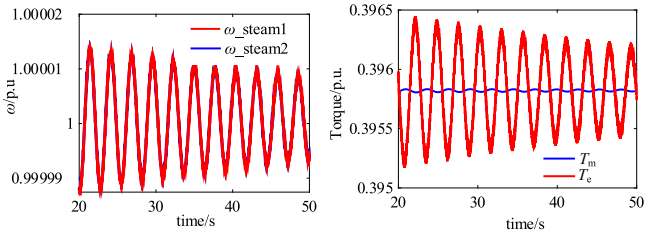


Fig. 29. Simulation results as $x_L = 0.02$H.

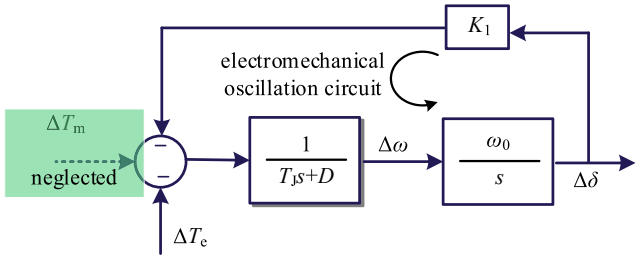


Fig. 30. Electromagnetic oscillation circuit when LFO occurs.

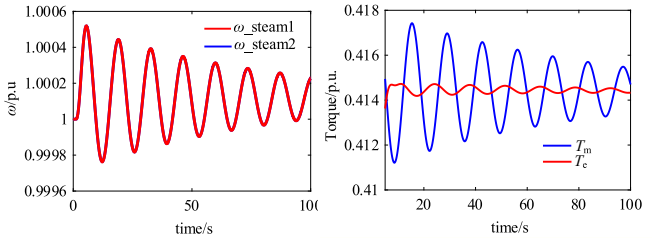


Fig. 31. Simulation results as $x_L = 20$H.

(2) In weak interconnected grid, the hydropower proportion will influence the damping level of system. The oscillation frequency and damping ratio decrease with increasing hydropower proportion, which indicates that a high hydropower proportion deteriorates the ULFO problem.
(3) In weak interconnected grid, the critical parameters of the governing system, including the steam and hydraulic governing parameters, as well as water hammer effect time constant of hydroturbine, significantly influence the ULFO.
(4) The essential difference between LFO and ULFO is related to the electromagnetic torque and mechanical torque, in ULFO, it is more likely to cause a considerable oscillation on mechanical torque, which will cause frequency oscillation.
(5) With higher penetration of renewable power units, the impact of renewable energy on ULFO should be studied in future research.

CRediT authorship contribution statement

Peng Wang: Conceptualization, Methodology. **Qin Jiang:** Methodology, Writing – original draft. **Baohong Li:** Formal analysis, Validation. **Tianqi Liu:** Investigation, Visualization. **Xiaopeng Li:** Resources, Supervision. **Gang Chen:** Resources, Supervision. **Xueyang Zeng:** Project administration. **Frede Blaabjerg:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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References

[1] Schleif F, White J. Damping for the northwest-southwest tie line oscillations-an analog study. IEEE Trans Power Appar Syst 1966;PAS-85(12):1239–47.
[2] Cebeci ME, Karaağaç U, Tör OB,. The effects of hydro power plants' governor settings on the stability of Turkish power system frequency. In: Proc. ELECO Conf., Bursa, Turkey, Dec. 2007. [Online]. Available: http://www.emo.org.tr/ekler/ ad6635f33710af6_ek.pdf.
[3] Pico HNV. Electromechanical oscillations in hydro-dominant power systems: an application to the Colombian power system. Ames, IA, USA: Dept. Elect. Eng., Iowa State Univ.; 2011. p. 10116. Ph.D. dissertation.
[4] Gao J, Li G, Wang K, Zhang H. Damping torque and synchronous torque analysis of power system ultra-low frequency oscillation. Power Syst Technol; 44(3): 1001–1007.
[5] He JB, Zhang JY, Li MJ, Li WF, Tang TN. Frequency domain analysis and control for governor stability problem in islanded HVDC sending systems. Proc CSEE 2013; 33(16):137–43.
[6] Mo WK, Chen YP, Chen HY, Liu YS. Analysis and measures of ultralow-frequency oscillations in a large-scale hydropower transmission system. IEEE J Emerg Sel Topics Power Electron 2018;6(3):1077–85.
[7] Chen G, Ding LJ, Li M. Stability characteristics of southwest China power grid after asynchronous interconnection. Power Syst Protec Contr 2018;46(7):76–82.
[8] Lu XM, Chen L, Chen YP. Ultra-low-frequency oscillation of power system primary frequency regulation. Autom Electr Power Syst 2017;41(16):64–70.
[9] Canay IM. A novel approach to the torsional interaction and electrical damping of the synchronous machine part I: theory. IEEE Trans Power Appar Syst 1982;PAS-101 (10):3630–3638.
[10] Liu CX, Zhang JF, Chen YP. Mechanism analysis and simulation on ultra low frequency oscillation of yunnan power grid in asynchronous interconnection mode. Southern Power Syst Technol 2016;10(7):29–34.

[11] Han XY, Jiang Q, Liu TQ, Li BH. Research on ultra-low frequency oscillation caused by hydro power in hydro-dominant power system. In: Proc Int Conf IEEE Power Syst Technol, Guangzhou, China, Nov. 2018. p. 1909–1914.
[12] Jiang Q, Li BH, Liu TQ. Investigation of hydro-governor parameters' impact on ULFO. IET Renew Power Gener 2019;13(16):3133–41.
[13] Cheng L, Wang WY, Xu G, Dou F. Damping analysis and parameter adjustment of hydraulic turbine governor under ultra-low frequency oscillation. In: Proc 3rd IEEE Conf Energy Int Syst Integ (EI2), Changsha, China, Apr. 2020. p.1424–1428.
[14] Gong TR, Wang GH, Li T. Analysis and control on ultra low frequency oscillation at seeding end of UHVDC power system. In: Proc Int Conf IEEE Power Syst Technol, Chengdu, China, Oct. 2014. p. 832–837.
[15] Teng YF et al. Mechanism and characteristics analysis of ultra-low frequency oscillation phenomenon in a power grid with a high proportion of hydropower. In: Proc Int Conf IEEE Power Syst Technol, Guangzhou, China, Nov. 2018. p. 575–584.
[16] Dandeno PL, Kundur P, Bayne JP. Hydraulic unit dynamic performance under normal and islanding conditions-analysis and validation. IEEE Trans Power Appar Syst 1978;PAS-97(6):2134–2143.
[17] Phi DT, Bourque EJ, Thorne DH, Hill EF. Analysis and application of the stability limits of a hydro-generating unit. IEEE Trans Power Appar Syst 1981;PAS-100(7): 3203–3212.
[18] Lu XM, Chen L, Zhang YW, Min Y, Wang MH. Limit cycle in the ultra-low-frequency oscillation of islanded power systems. In: Proc IEEE Int Conf Contr Autom (ICCA), Ohrid, Macedonia, Jul. 2017. p. 648–653.
[19] Pico HV, McCalley JD, Angel A, Leon R, Castrillon NJ. Analysis of very low frequency oscillations in hydro-dominant power systems using multi-unit modeling. IEEE Trans Power Syst 2012;27(4):1906–15.
[20] Yang WJ, et al. Experimental investigation of theoretical stability regions for ultra-low frequency oscillations of hydropower generating systems. Energy 2019;186(1): 115816.
[21] Wang GT, Xu Z, Guo XY, Xiao L. Mechanism analysis and suppression method of ultra-low-frequency oscillations caused by hydropower units. Int J Electr Power Energy Syst 2018;103(1):102–14.
[22] Jiang CX, Zhou JH, Shi P, Huang W, Gan DQ. Ultra-low frequency oscillation analysis and robust fixed order control design. Int J Electr Power Energy Syst 2019; 104:269–78.
[23] Zheng C et al. Analysis and control to the ultra-low frequency oscillation in southwest power grid of China: a case study. In: Proc Chinese Contr Decis Conf (CCDC), Shenyang, China, Jun. 2018. p. 5721–5724.
[24] Brezovec M, Kuzle I, Krpan M, et al. Analysis and treatment of power oscillations in hydropower plant Dubrava. IET Renew Power Gener 2020;14(1):80–9.
[25] Brezovec M, Kuzle I, Krpan M, et al. Improved dynamic model of a bulb turbine-generator for analysing oscillations caused by mechanical torque disturbance on a runner blade. Int J Electr Power Energy Syst 2020;119:1–9.
[26] Yang W, Norrlund P, Bladh J, et al. Hydraulic damping mechanism of low frequency oscillations in power systems: quantitative analysis using a nonlinear model of hydropower plants. Appl Energy 2018;212:1138–52.
[27] Yang W, Norrlund P, Chung C, et al. Eigen-analysis of hydraulic-mechanical-electrical coupling mechanism for small signal stability of hydropower plant. Renew Energy 2018;115:1014–25.
[28] Jiang ZS, Sun LX, Zhou HQ, et al. The dynamic system of the thermal power impact on the ultra-low frequency oscillation analysis. In: Proceedings of the 24th academic annual conference of electric power system and automation in Chinese Universities, Beijing, Nov. 2008.
[29] Chen L, Lu XM, Chen YP, Min Y, Zhang Y, Tang ZY. Analysis of ultra-low frequency oscillations in multi-machine system and equivalent method. Autom Electr Power Syst 2017;41(22):10–5.
[30] Kundur P. Power system stability and control. New York, NY, USA: McGraw-Hill; 1994.
[31] Wang P, Li B, Zhao J, Liu T, Jiang Q, Chen G. The mechanism of ultra-low frequency oscillations with the same mode shapes. IEEE Access 2020:198047–57.